

Nirma University

Institute of Technology

Semester End Examination (IR), December-2025

Master of Computer Application, Semester - 1

6CS102CC22/6CS102CC25 Data Structures

Roll/

Exam No.

Time: 3 Hours

Supervisor's Initial
with date

Max Marks: 100

Instructions:

1. Attempt all questions. (No Optional Questions)
2. Use section-wise separate answer book.
3. Figures to the right indicate full marks.
4. Draw neat sketches wherever necessary.
5. Assume suitable data wherever necessary and mention the same.

SECTION - I

Q-1. Answer the following questions.

[16]

(A) What is the meaning of the complexity of an algorithm? Explain (4)
CO1 BL2 worst case, average case and best case with respect to the algorithm complexity.

(B) For array *Ary* of dimension 4X4 with base address 4000 and (6)
CO1 BL3 element memory size of 4 bytes, find the address of element *Ary*[3][3] using row-major order and column-major order. Consider the lower bound for rows and columns is 0.

(C) Represent the polynomial given below using a singly linked list and (6)
CO1 BL3 explain the node format for the same. Write an algorithm to count the number of terms in this polynomial.

Polynomial: $43X^3Y^2Z^3 + 50X^2Y^3Z + 15XYZ + 15X + 40Y^2 + 8Z + 6$

Q-2. Answer the following questions.

[18]

(A) Convert the following infix expression into a postfix expression. (6)
CO2 BL3 Solve the resultant postfix expression using a stack.
 $2 + 3 * (9 - 4) / 5$

(B) The list prepared using dynamic memory allocation is flexible in (6)
CO4 BL5 size. Justify this statement with appropriate examples. Also, write an algorithm to add a customer at the end of the list. Create this list using dynamic memory allocation.

(C) Develop a patient registration system using a suitable data (6)
CO4 BL6 structure. The system allows adding new patients, calling the next patient, and viewing the waiting list. Formulate algorithms to simulate this system. Consider that the system uses static memory allocation.

Q-3. Answer the following questions.

[16]

(A) Perform a binary search method on the list of values given below to search for the value 40. Also, compare linear search and binary search in terms of time complexity.

CO3 BL4

(4)

15 30 36 40 52 69 71 83 98

(B) Show tracing to sort the following data in ascending order using Selection Sort:

CO3 BL3

(6)

44 31 75 24 82 22 17 28

(C) Write algorithm(s) to perform Merge Sort. Discuss the time complexity for the same.

CO3 BL3

(6)

SECTION – II

Q-4. Answer the following questions.

[16]

(A) Differentiate the following with respect to a graph.

CO1 BL4

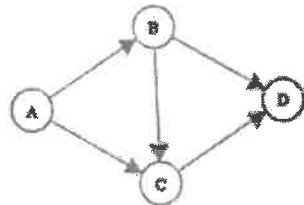
(6)

- (i) Mixed graph vs Multigraph
- (ii) Weighted graph vs Null graph
- (iii) Simple path vs Elementary path

(B) What do you mean by path matrix? Write its usage. Find the path matrix for the given graph below.

CO2 BL3

(10)



Q-5. Answer the following questions.

[18]

(A) Construct a Binary Tree for the given in-order and post-order traversals.

CO2 BL3

(6)

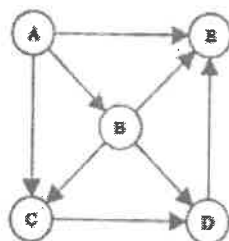
Post-order: D F E B G L M J K H C A

In-order: D B F E A G C L J M H K

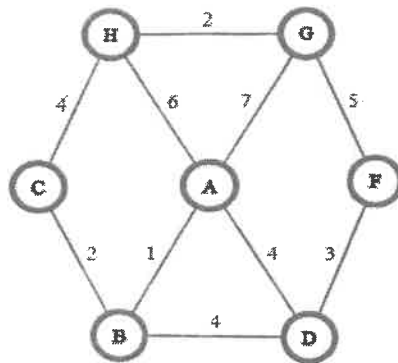
(B) Represent the following digraph using an Adjacency matrix. Find the in-degree, out-degree and total degree of each vertex.

CO2 BL3

(6)



- (C) For the given graph below, find the minimum spanning tree and the total cost of the minimum spanning tree using Kruskal's algorithm. Show all the intermediate steps. (6)
- CO2 BL3



Q-6. Answer the following questions.

[16]

- (A) Create a Binary Search Tree (BST) for the following data. Write the in-order traversal, pre-order traversal and post-order traversal for the generated BST. (8)
- CO3 BL3

41 22 65 11 34 74 9 38

- (B) Differentiate between B-Tree and B+Tree. Construct a B-tree of order 3 by inserting the following numbers in the given sequence. (8)
- CO3 BL4

14 25 8 10 17 21 47 3 6 9 7 5